The documentation and process conversion measures necessary to comply with this amendment shall be completed by 29 November 1994

INCH-POUND

MIL-S-19500/441D
AMENDMENT 1
29 May 1996

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, POWER
TYPES 2N3740 AND 2N3741,
JANTX, JANTXV, JANS, AND JANC

This amendment forms a part of MIL-S-19500/441D, dated 3 May 1993, and is approved for use by all Departments and Agencies of the Department of Defense

PAGE 3

FIGURE 1, Dimensions table, ϕp, notes column; delete "8" and substitute "7".

PAGE 7

4.4.1, delete and substitute as follows:

"4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. End-point electrical measurements shall be in accordance with table I, subgroup 2 herein."

4.4.2, delete and substitute as follows:

"4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2 herein."

4.4.2.2, delete and substitute as follows:

"4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500. Subgroup 3: Intermittent operation life (LTPD); method 1037; $V_{CB} \geq 10$ V dc, ΔT_J between cycles $\geq +100°C$, $t_{on} = t_{off} \geq 1$ minute for 2,000 cycles. No heat sink or forced air cooling on the device is permitted."

4.4.3, delete and substitute as follows:

"4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2 herein."

4.4.3.1, subgroup 6, delete and substitute as follows:

"b. Intermittent operation life (LTPD); method 1037; $V_{CB} \geq 10$ V dc, ΔT_J between cycles $\geq +100°C$, $t_{on} = t_{off} \geq 1$ minute for 6,000 cycles. No heat sink or forced air cooling on the device is permitted."

4.4.4, delete in its entirety.

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
13	13
14	REPRINTED WITHOUT CHANGE

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PAGE 8

TABLE I, subgroup 2, conditions column: Add "2/ 3/ 4/".

PAGES 10 and 11

TABLE I, subgroup 5, end-point electrical measurements, conditions column, two places, delete and substitute the following:

"See table I, subgroup II herein."

PAGE 11

TABLE I, add the following footnotes:

"2/ Delta requirements for JANS, table IVa of MIL-S-19500 are as follows:
a. Subgroup 4, Δh_{FE2} = ±25 percent, $\Delta V_{CE(sat)1}$ = ±50 mV.
b. Subgroup 5, ΔI_{CEX1} = 100 percent of initial value or 50 nA dc, whichever is greater. Δh_{FE2} = ±25 percent, $\Delta V_{CE(sat)1}$ = ± 50 mV.

3/ Delta requirements for JANTX and JANTXV, table IVb of MIL-S-19500 are as follows:. Subgroup 6, Δh_{FE2} = ±25 percent.

4/ Delta requirements for JANS, table V of MIL-S-19500 are as follows: Subgroup 6, Δh_{FE2} = ±25 percent, ΔI_{CEX1} = 100 percent of initial value or 50 nA dc, whichever is greater."

TABLE II: Delete in its entirety.

PAGE 12

TABLE III: Delete in its entirety.

PAGE 17

6.2b, delete and substitute as follows:

"b. Lead finish as specified (see 3.3.1)."

PAGE 18

CONCLUDING MATERIAL: Delete and substitute as printed below.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

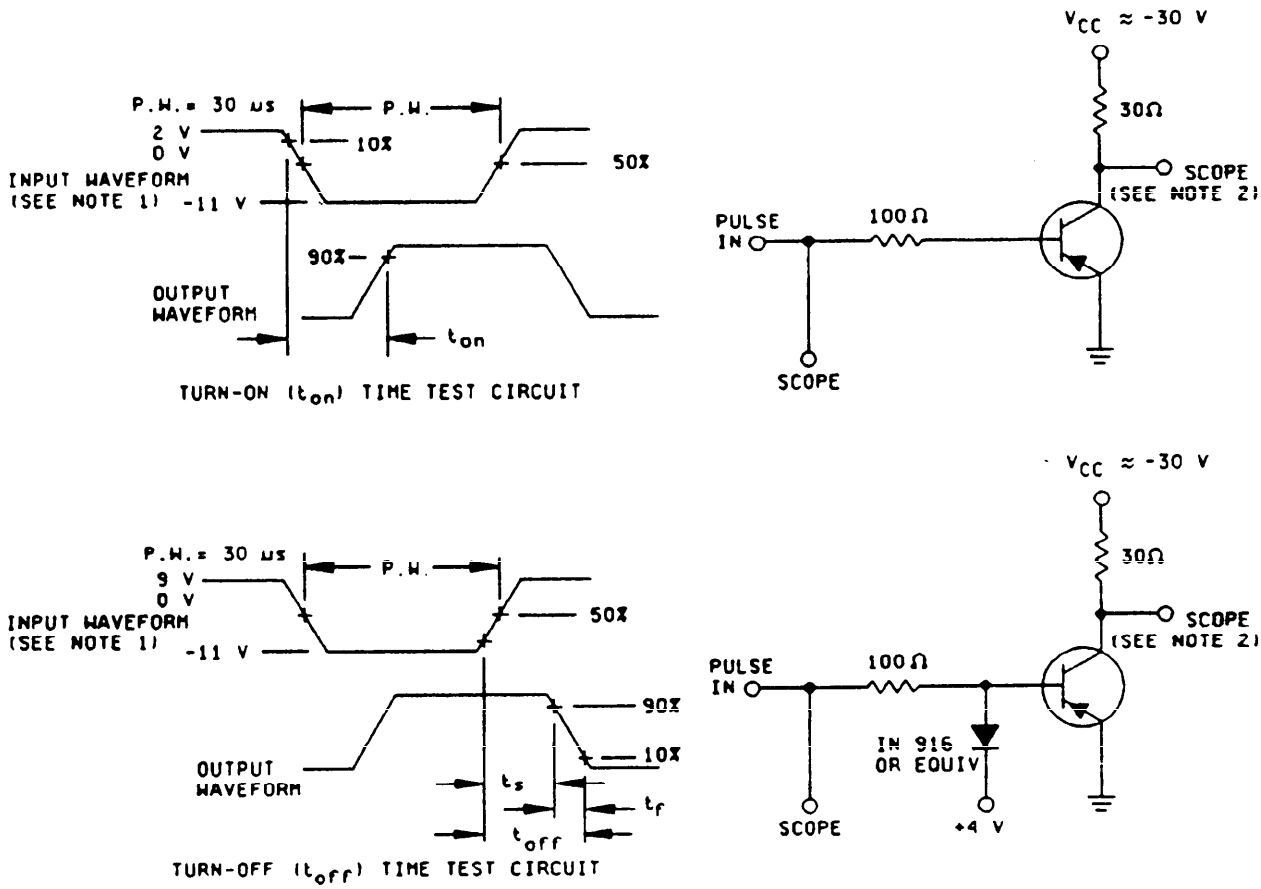
Review activities:
Army - AR, AV, MI, SM
Navy - AS, CG, MC
Air Force - 13, 15, 19, 85, 99
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 5961-1562)

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NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 20 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50Ω.
2. Output sampling oscilloscope: Z_{in} ≥ 100 kΩ; C_{in} ≤ 12 pF; rise time ≤ 2 ns.

FIGURE 3. Pulse response test circuits.

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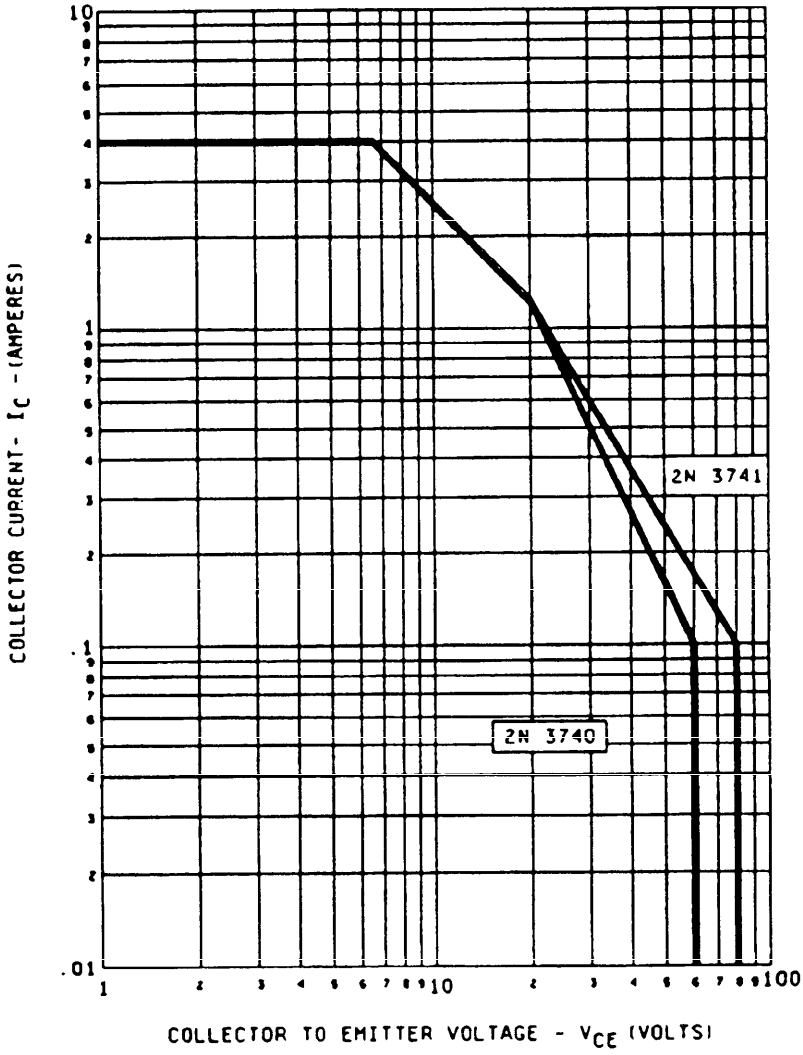


FIGURE 4. Maximum safe operating area graph (continuous dc).

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